

DATABASE MANAGEMENT
SYSTEM
CSA0503
TECHNICAL PRESENTATION

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Question; Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.

Introduction

File-Based System

- Stores data in separate files.
- Each department maintains its own records.
- Data sharing is difficult.
- Suitable for small organizations.

Database Management System (DBMS)

- Stores data in a centralized database.
- Allows multiple users to access data.
- Provides security, backup, and faster data retrieval.
- Suitable for medium and large organizations.

Comparison Between File-Based System and DBMS

Feature	File-Based System	DBMS
Data Storage	Separate files	Centralized database
Data Redundancy	High	Low
Data Sharing	Difficult	Easy
Security	Low	High
Data Retrieval	Slow	Fast
Backup & Recovery	Manual	Automatic
Multi-user Access	Not Supported	Supported

Banking Case Study – File-Based System Scenario

- ▶ A small bank stores customer accounts, loans, and transactions in separate files.

Problems

- ▶ Duplicate customer records.
- ▶ Slow account updates.
- ▶ Errors during transactions.
- ▶ Difficult to generate reports.
- ▶ Increased maintenance cost.

Result

- ▶ Poor customer service.
- ▶ Higher chances of data inconsistency.

Banking Case Study – DBMS

Solution

- ▶ The bank uses a centralized DBMS for all customer information.

Benefits

- ▶ Real-time account updates.
- ▶ Secure online and ATM transactions.
- ▶ Faster loan processing.
- ▶ Accurate customer records.
- ▶ Better fraud detection.

Outcome

- ▶ Improved customer satisfaction.
- ▶ Faster banking operations.

Healthcare Case Study

File-Based System Problems

- ▶ Patient records stored in separate files.
- ▶ Duplicate medical records.
- ▶ Delayed access during emergencies.
- ▶ Billing errors.

DBMS Solution

- ▶ Centralized Electronic Health Records (EHR).
- ▶ Instant access to patient history.
- ▶ Accurate billing and prescriptions.
- ▶ Better coordination between departments.

Outcome

- ▶ Faster treatment.

Advantages and Disadvantages

File-Based System

Advantages

- ▶ Simple to use.
- ▶ Low initial cost.

Disadvantages

- ▶ Data redundancy.
- ▶ Poor security.
- ▶ Slow retrieval.
- ▶ Difficult maintenance.

DBMS

Advantages

- ▶ High security.
- ▶ Fast data access.
- ▶ Reduced redundancy.
- ▶ Easy backup and recovery.

Disadvantages

- ▶ Higher installation cost.
- ▶ Requires trained administrators.

Real-Life Applications

Banking

- ▶ Internet Banking
- ▶ ATM Transactions
- ▶ Loan Management
- ▶ Customer Account Management

▶ **Healthcare**

- ▶ Electronic Health Records (EHR)
- ▶ Hospital Billing
- ▶ Pharmacy Management
- ▶ Laboratory Information Systems
- ▶ Appointment Scheduling

Conclusion

- ▶ File-based systems are suitable for small organizations with limited data.
- ▶ DBMS offers centralized, secure, and efficient data management.
- ▶ Banking and healthcare sectors depend on DBMS for reliable and accurate operations.
- ▶ Adopting DBMS improves productivity, reduces errors, and enhances customer and patient services.